# Mobile application Bluetooth frame documentation

July 22, 2021

## 1 Bluetooth data format

CANlink wireless series 3000 uses the RM Byte Command Protocol, which serves for bi-directional transparent CAN message transmission between the host and the CAN network. To transfer CAN message through Bluetooth it needs to be encoded into command frames with specific byte values.

| Byte(s) | Value | Description |
|---|---|---|
| **SOF (Start of Frame)** | 'C' | SOF marks the beginning of the command. 'C' (0x43) is the default setting. For some devices this value can be changed. This manual always shows the default setting. |
| **Length** | 0x00-0xFF | The length byte includes the number of data bytes + the number of command bytes following. |
| **Command** | 0x00-0xFF | Command byte, values 0x00 to 0xFE are direct commands, the value 0xFF means that the next byte contains a command extension |
| **Data** | xx | Data content |
| **Checksum** | xx | Contains the XOR checksum of the SOF, Length, Command and the Data-bytes. |
| **EOF (End of Frame)** | <CR> | EOF byte marks the end of the command. 0x0D (Carriage Return) is the default setting. For some devices this value can be changed. This manual always shows the default setting |

"--" means that this value is not of interest
"xx" means that the field contains the configuration / reply value

## 2 Sensor data info Bluetooth frames

### 2.1 Engine frame 1

From TTControl

- Accumulator voltage (ACC_VOL)
- Fuel Temperature (FT)
- Fuel Level (in litres) (FL)
- Engine Fuel Rate (EFR)
- Torque of engine (TRQ)
- Engine Rotations Per Minute (RPM)

**Frame length = 8 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x64 | RPM_1 | RPM_0 | TRQ | EFR | FL | FT | ACC_VOL_1 | ACC_VOL_0 |

Decoding:

- $ACC_VOL = ((ACC_VOL_1 << 8)|(ACC_VOL_0))/1000$
- $EFR = EFR/10$

### 2.2 Engine frame 2

From TTControl

- ETH - engine total hours
- EFT - engine fuel total

**Frame length = 8 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x65 | ETH_3 | ETH_2 | ETH_1 | ETH_0 | EFT_3 | EFT_2 | EFT_1 | EFT_0 |

Decoding:

- $ETH = ((ETH_1 << 8)|(ETH_0))/10$
- $EFT = ((EFT_1 << 8)|(EFT_0))/10$

### 2.3 Hydraulics data

From TTControl

- Momentary predicted work time with remaining fuel (WT)
- Hydraulic oil temperature (OT - usint)
- Hydraulic oil pressure (PRS)

**Frame length = 8 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x69 | 0xFF | WT | OT | PRS_1 | PRS_0 | RNG | 0xFF | 0xFF |

Decoding:

- $OT = OT - 80$

### 2.4 Carts angle and chassis angle

From TTControl

- CR_F - front cart rotation (CR=CR/10)
- CR_B - back cart rotation (CR=CR/10)
- X axis inclinometer value (along Agribot chasis), (X_INC=X_INC/10)
- Y axis inclinometer value (along Agribot chasis), (Y_INC=Y_INC/10)

**Frame length = 8 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x6A | CRF_1 | CRF_0 | CRB_1 | CRB_0 | X_INC_1 | X_INC_0 | Y_INC_1 | Y_INC_0 |

Decoding:

- All above values needs to be devided by 10 (same maner as in ETH in Frame 2).

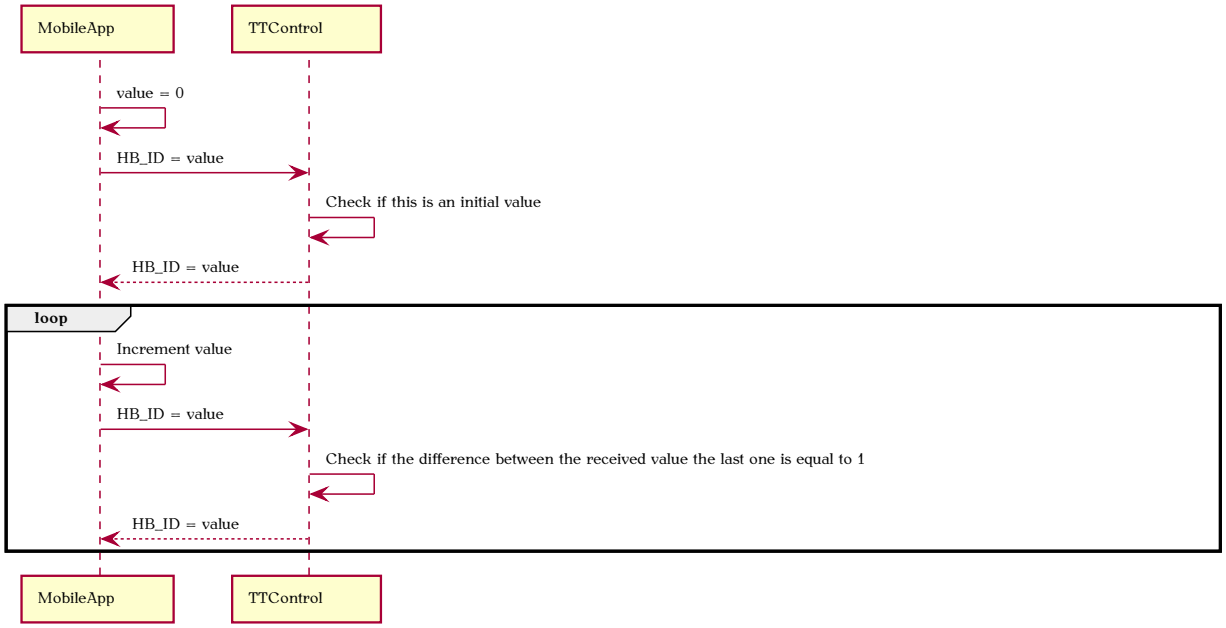
# 3 State changing and command frames

## 3.1 Heartbeat frame

From Mobile Application and TTControl. This frame needs to be send with proper value at least every 500 ms or robot will terminate the connection with mobile application. The initial Heartbeat value (HB_ID) of 0 is send by mobile application and then robot answers with the same value of HB_ID, which is then incremented by the application and send back. Robot once again check if the HB_ID is proper and send back the answer or breaks the connection if the value was bad. This process is repeated cyclically (at least every 200 ms), every 255 the HB_ID rounds back at 0.

**Frame length = 1 byte**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x01 | HB_ID | * | * | * | * | * | * | * |




## 3.2 Handshake frame

From Mobile Application. This frame initializes all the communication with the robot.

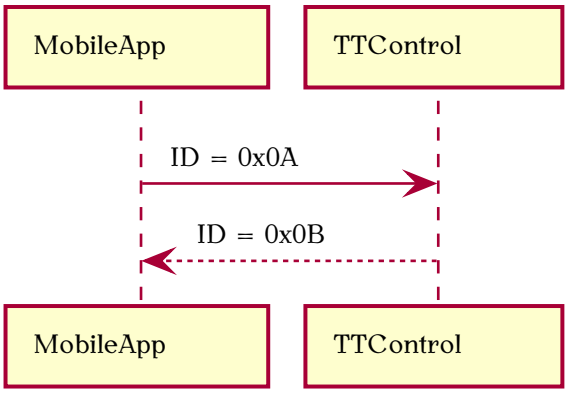
**Frame length = 1 byte**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x0A | 0x01 | * | * | * | * | * | * | * |

**Frame length = 1 byte**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x0B | 0x01 | * | * | * | * | * | * | * |

TTControl send the exact same message as the confirmation of receiving this command.




### 3.3 Manual steering initialization

From Mobile Application. The data is only one dummy byte. This frame is used to initiate the manual steering state.

**Frame length = 1 byte**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x14 | 0x14 | * | * | * | * | * | * | * |

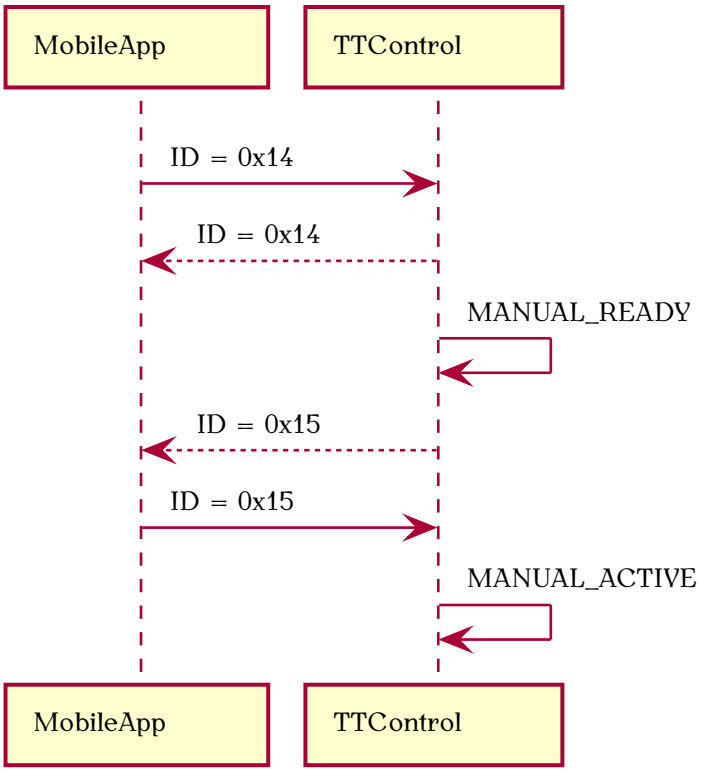
TTControl send the exact same message as the confirmation of receiving this command.

### 3.4 Manual steering ready

From TTControl. The data is only one dummy byte. This frame is send to inform mobile application that the manual steering commands set is now active.

**Frame length = 1 byte**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x15 | 0x15 | * | * | * | * | * | * | * |


MobileApp
TTControl
ID = 0x14
ID = 0x14
MANUAL_READY
ID = 0x15
ID = 0x15
MANUAL_ACTIVE
MobileApp
TTControl

# 4 Error frames

## 4.1 Error frames

If the there is an error with robot, a frame describing the problem is send

- ERR – a 2-byte number describing error code

**Frame length = 2**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0xC8 | ERR_1 | ERR_0 | * | * | * | * | * | * |

## 4.2 Engine error frames start

If there are errors in the engine, a frame announcing the problem is send

- ERR_TOT – a byte number describing total error count
- LAMP_STATE – the state of lamps which describe the severity of the problems

**Frame length = 2**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x104 | ERR_TOT | LAMP_STATE | * | * | * | * | * | * |

## 4.3 Engine error frames

A frame containing one of the announced error frames. Byte order LSB first.

- ERR_NUM – an number number, value up to the announced in Engine error frames start
- SPN – the 3-byte error code described in Deutz documents
- FMI – the error specific code described in Deutz documents
- OCC – the error occurrence count
- ERR_STATE – if the error is active, contains 1, if passive 0

**Frame length = 8**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x105 | ERR_NUM | SPN_0 | SPN_1 | SPN_2 | FMI | OCC | ERR_STATE | 0xFF |

## 4.4 Manual error frames

If the there is an error during manual commands, an error frame is sent

- ERR – a 2-byte number describing error code

**Frame length = 2**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0xC9 | ERR_1 | ERR_0 | * | * | * | * | * | * |

# 5 Manual command frames

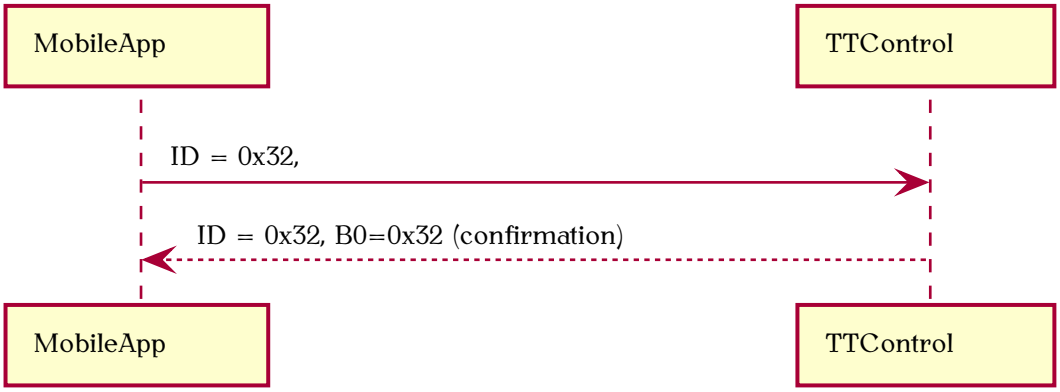
## 5.1 Override cart linear movement

Mobile application to TTControl.

- DIR - if 1, go forward, if 0, go backward
- CART_VEL - velocity of cart (in old agribot default value was 0xFF)
- FRONT_SEL - if 1, front cart selected. If 0, front cart deselected
- BACK_SEL - if 1, back cart selected. If 0, back cart deselected

**Frame length = 4**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x32 | DIR | CART_VEL | FRONT_SEL | FRONT_SEL | * | * | * | * |



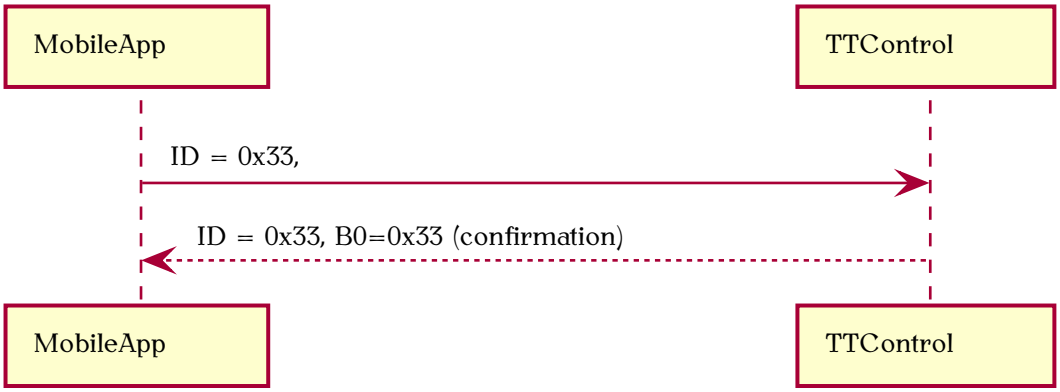

## 5.2 Override cart rotation

Mobile application to TTControl.

- CART_FRONT - if 1, front cart is selected and will rotate, if 0, front cart will be ignored
- DIR_FRONT - if 1, rotate front cart left, if 0, rotate front cart right
- CART_BACK - if 1, back cart is selected and will rotate, if 0, back cart will be ignored
- DIR_BACK - if 1, rotate back cart left, if 0, rotate back cart right

**Frame length = 4**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x33 | CART_FRONT | DIR_FRONT | CART_BACK | DIR_BACK | * | * | * | * |



### 5.3 Joystick frame

From Mobile Application. This frame is used to move the robot around and needs to be send every 300 ms or the command timeout will occur and the controller will halt the outputs and the robot will stop.

- DIR_V - velocity direction (negative velocity means forward movement, positive - backward movement)
- DIR_R - radius direction (negative radius means right, positive - left)
- RAD - radius value
- VEL - velocity value

**Frame length = 8**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x35 | DIR_V | DIR_R | RAD_1 | RAD_0 | VEL_1 | VEL_0 | 0xFF | 0xFF |

Decoding:

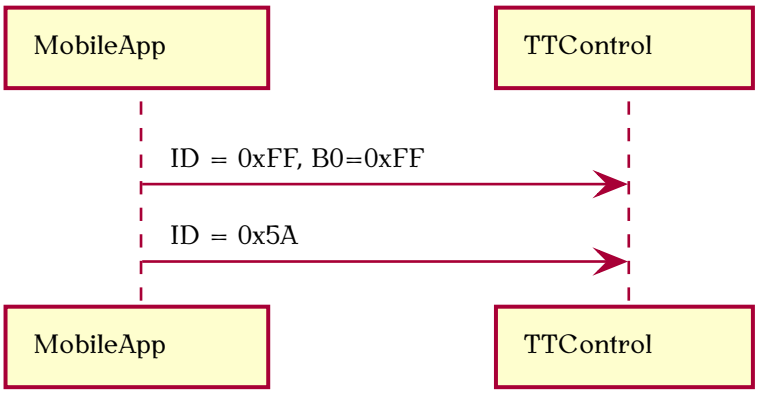
- $RAD = RAD/10$
- $VEL = VEL/100$

### 5.4 Start Engine

From Mobile Application. This frame is used to start the engine, by sending 2 of them in interval shorter than 200 ms and stop the engine by sending it for 2 seconds.

**Frame length = 1**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x5A | 0xFF | * | * | * | * | * | * | * |

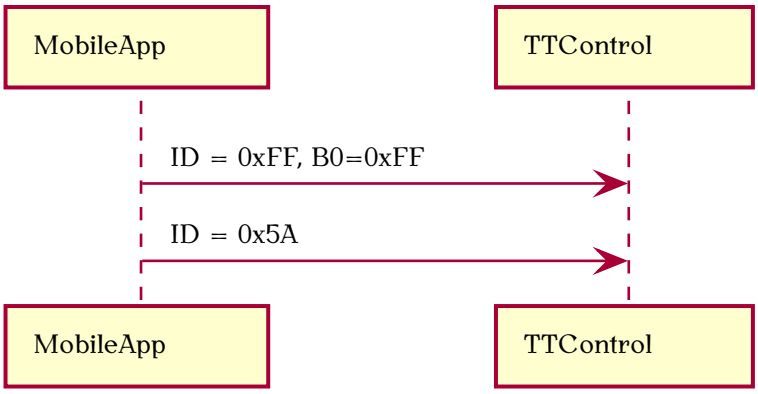



### 5.5 STOP Engine

From Mobile Application. This frame is used to stop any ongoing track moving tasks like, aligning carts and switches. It is used to start and stop the engine too, by sending first this frame and then an Start Stop engine frame.

**Frame length = 1**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0xFF | 0xFF | * | * | * | * | * | * | * |




### 5.6 Change front delta geometry

From Mobile Application. This is a request frame used to put up or down front caterpillars.

- REQ_RES - a code number describing the meaning of the frame (request or response), in this case 0 (change request)
- CAT_STA - caterpillar state, 0 for normal caterpillar state, 1 (anything but 0) for elevated caterpillar state

**Frame length = 2 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F5 | REQ_RES | CAT_STA | * | * | * | * | * | * |

From Advantech to Mobile Application. This frame is a callback from putting up or down front caterpillars.

- REQ_RES - a code number describing the meaning of the frame (request or response), in this case 1 (change response)
- CAT_STA - caterpillar state, 0 for normal caterpillar state, 1 for elevated caterpillar state, 2 for unknown caterpillar state

**Frame length = 2 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F5 | REQ_RES | CAT_STA | * | * | * | * | * | * |

### 5.7 Change back delta geometry

From Mobile Application. This frame is used to put up or down back caterpillars.

- REQ_RES - a code number describing the meaning of the frame (request or response), in this case 0 (change request)
- CAT_STA - caterpillar state, 0 for normal caterpillar state, 1 (anything but 0) for elevated caterpillar state

**Frame length = 2 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F6 | REQ_RES | CAT_STA | * | * | * | * | * | * |

From Advantech to Mobile Application. This frame is a callback from putting up or down back caterpillars.

- REQ_RES - a code number describing the meaning of the frame (request or response), in this case 1 (change response)

- CAT_STA - caterpillar state, 0 for normal caterpillar state, 1 for elevated caterpillar state, 2 for unknown caterpillar state

**Frame length = 2 bytes**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F6 | REQ_RES | CAT_STA | * | * | * | * | * | * |

# 6 Tools control

## 6.1 Control tool override - controlling the valves

From Mobile Application to TTControl (also from Advantech to TTControl). Every switch control a hydraulic valve connected to a specific part of the robot tools.

### 6.1.1 Frame

- SW_ID - switch ID number
- SW_ON - the switch should be ON (1) or OFF (0)
- SW_LV - velocity value a percent integer number from 0 to 100, numbers larger than 100 are treated as mistakes and the switch is instantly closed

**Frame length = 3**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x3C | SW_ID | SW_ON | SW_LV | * | * | * | * | * |

### 6.1.2 SW_ID

- 0x00 - Config to all switches
- 0x01 - FRONT_PTO
- 0x02 - FRONT_RED
- 0x03 - FRONT_GREEN
- 0x04 - FRONT_BLUE
- 0x05 - FRONT_TPL_ANGLE
- 0x06 - FRONT_TPL_HEIGHT
- 0x07 - BACK_PTO
- 0x08 - BACK_RED
- 0x09 - BACK_GREEN
- 0x0A - BACK_BLUE
- 0x0B - BACK_TPL_ANGLE
- 0x0C - BACK_TPL_HEIGHT

In return TTControl sends back confirmation frame:

**Frame length = 1**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x3C | 0x3C | * | * | * | * | * | * | * |

## 6.2 Control tool position - manual mode steering

From Mobile Application to Advantech. In this case, the destination configuration of tool is sent via the CAN frame. The units inside the frame differs depending on SW_ID. Details are listed in below subsections.

### 6.2.1 Frame - set TPL position - request

- CONF_ID - config ID number (IDs are identical to those listed in section [6.1.2])
- CONF_VAL_Bx - values specific to ID

**Frame length = 3**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x13D | CONF_ID | CONF_VAL_B1 | CONF_VAL_B0 | * | * | * | * | * |

### 6.2.2 Frame - set TPL position response

From Advantech to Mobile app.

- CONF_ID - config ID number (IDs are identical to those listed in section [6.1.2])
- RET - 0xFF - failed to set TPL config (for instance values are out of range); 0x00 - successfully set TPL config

**Frame length = 2**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x13E | CONF_ID | RET | * | * | * | * | * | * |

### 6.2.3 CONF_VAL_Bx units and frame decoding

- FRONT_PTO\BACK_PTO - rotations per minute. Range from 0 RPM to 1000 RPM. The value in the frame is delivered on two bytes. The value can be calculated using following formula:
  $VAL = (CONF_VAL_B1 << 8)|(CONF_VAL_B0)$
- FRONT_GREEN\BACK_GREEN - valve flow control. Range from 0 to 100%.
  $VAL = (CONF_VAL_B0)$
- FRONT_YELLOW\BACK_YELLOW - valve flow control. Range from 0 to 100%.
  $VAL = (CONF_VAL_B0)$
- FRONT_BLUE\BACK_BLUE - valve flow from 0 to 100.
  $VAL = (CONF_VAL_B0)$
- FRONT_TPL_ANGLE\BACK_TPL_ANGLE - radians (from -0.26rad to 0.52rad). The negative number can be achieved by setting the highest bit to 1 in CONF_VAL_B1. The decoding can be performed by extracting sign bit, then setting the sign bit to 0. The resulding number should be devided by 1000.

  Example decoding: 0x8521. The 0x80 is indicates that the number is negative. We set the highest bit to 0. 0x0521 in hex is 1313. We divide the number by 1000 and multiply the number by -1. The result is -0.1313
- FRONT_TPL_HEIGHT\BACK_TPL_HEIGHT - ejection of prismatic joint (from 0.0m to 0.45m). The decoding process is analogous to to angle frame, but without the sign

## 6.3 TPL position feedback

Frame send periodically from Advantech to Mobile app. Feedback from sensors mounted on robot.

- CONF_ID - config ID number (IDs are identical to those listed in section [6.1.2])
- CONF_VAL_Bx - values specific to ID

**Frame length = 3**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x13F | CONF_ID | CONF_VAL_B1 | CONF_VAL_B0 | * | * | * | * | * |

# 7 RFID frames

## 7.1 Trigger scan

From mobile application to Advantech. The frame triggers scan from selected antenna.

- COM – a code number describing the meaning of the frame, in this case 1 (scan request)
- ANT_SEL – 0x01 - front antenna selected, 0x02 - back antenna selected, 0x03 - both antennas selected

**Frame length = 2**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F4 | COM | ANT_SEL | * | * | * | * | * | * |

## 7.2 Detected tags - response

From Advantech to mobile application. In reply to trigger scan frame [7.1], the tags ID are sent. The EPC can be long up to 496 bits. In following example, the EPC tag is 9 bytes long.

- COM – a code number describing the meaning of the frame, in this case 2 (scan response start)
- TAG_ANT – 0x01 - tag detected by front antenna, 0x02 - tag detected by back antenna
- TAG_LEN – numbers of bytes
- TAG_ID – a 2-byte number representing the RFID tag ID

**Frame length = 3**

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F4 | COM | TAG_ANT | TAG_LEN | * | * | * | * | * |

**Frame length = 8**

- COM – a code number describing the meaning of the frame, in this case 3 (scan response tag data pt 1)

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F4 | COM | TAG_ID_B0 | TAG_ID_B1 | TAG_ID_B2 | TAG_ID_B3 | TAG_ID_B4 | TAG_ID_B5 | TAG_ID_B6 |

**Frame length = 8**

If the ID is larger than capacity of frame, COM will increment.

- COM – a code number describing the meaning of the frame, in this case 4 (scan response tag data pt 2)

| ID | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7 |
|---|---|---|---|---|---|---|---|---|
| 0x1F4 | COM | TAG_ID_B8 | TAG_ID_B9 | 0x00 | 0x00 | 0x00 | 0x00 | 0x00 |